

When the exposure looks exactly like the gene

Toxic and pharmacologic exposures that reproduce a Mendelian phenotype closely enough to mislead a phenotype-driven pipeline. **Deeper phenotyping does not resolve these — it makes the wrong gene rank higher.** What resolves them is the exposure history and a specific non-genetic test.

Compiled for monarch-initiative/dismech#10395 · 2026-09-02 · 48 pairs, 73 genes, 15 organ systems

48 EXPOSURE-GENE PAIRS	13 TIER 1 · IDENTICAL MOLECULAR TARGET	73 DISTINCT GENES OR LOCI	15 ORGAN SYSTEMS REPRESENTED	20 HALT OR REGRESS ON WITHDRAWAL
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How to read the tier column

- T1 Identical molecular target.** The agent inhibits the exact gene product. Downstream of that node there is no biochemical difference to find — no metabolic panel separates these, and no amount of additional phenotyping will.
- T2 Shared pathway or organ mechanism, different node.**
- T3 Convergent phenotype, no shared molecular lesion.**

CLASS A Identical molecular target14 pairs

The agent inhibits the very protein the Mendelian disease disrupts. Downstream of that node there is no biochemical difference to find — these are the pairs a metabolic panel cannot separate.

AGENT / EXPOSURE	MENDELIAN MIMIC	SHARED LESION	WHAT SEPARATES THEM
T1 Licorice / glycyrrhizin confectionery, herbal tea, smokeless tobacco	Apparent mineralocorticoid excess reversible <i>HSD11B2</i> Endocrine	11β-HSD2 inhibited; cortisol not converted to cortisone	Dietary history; cortisol:cortisone ratio normalises weeks after withdrawal
T1 Lead paint, ammunition, informal recycling	ALAD-deficiency porphyria (Doss); sideroblastic anaemia reversible <i>ALAD</i> · <i>ALAS2</i> Haematologic	Zn ²⁺ displaced from ALA dehydratase	Blood lead; ↑ urinary ALA with normal PBG; basophilic stippling; chelation response

AGENT / EXPOSURE	MENDELIAN MIMIC	SHARED LESION	WHAT SEPARATES THEM
T1 Organophosphates, TOCP pesticides, adulterated oils, jet lubricants	SPG39 · Boucher-Neuhäuser · Gordon Holmes reversible <i>PNPLA6</i> Neurologic (PNS)	Neuropathy target esterase inhibited then aged	Exposure history; onset 1–3 weeks post-exposure; static or improving course
T1 Nitrous oxide recreational canisters, dental and obstetric use	Functional cblG / remethylation defect reversible <i>MTR · MTRR</i> Neurologic (CNS)	Cobalamin oxidised; methionine synthase inactivated	Use history; ↑tHcy and ↑MMA correct with hydroxocobalamin; normal sequence
T1 Fluconazole, high dose first-trimester antifungal therapy	Antley-Bixler syndrome <i>POR · FGFR2</i> Skeletal	CYP51A1 (lanosterol 14α-demethylase) inhibited	Maternal exposure timing; normal steroid profile and sterol intermediates
T1 Aripiprazole, trazodone, haloperidol routine psychotropic prescribing	Smith-Lemli-Opitz (biochemical) reversible <i>DHCR7</i> Metabolic	DHCR7 inhibited; 7-dehydrocholesterol accumulates	7-DHC falls on washout; no growth failure, 2–3 toe syndactyly or dysmorphism
T1 Hypoglycin A unripe ackee, sycamore maple seeds	MCAD deficiency · MADD reversible <i>ACADM · ETFDH</i> Metabolic	MCPA-CoA inhibits medium-chain acyl-CoA dehydrogenase	MCPA-carnitine species on the acylcarnitine profile; dietary history
T1 Vigabatrin infantile spasms, refractory focal epilepsy	GABA-transaminase deficiency; Leigh-like MRI reversible <i>ABAT · SURF1</i> Neurologic (CNS)	ABAT irreversibly inhibited; symmetric T2 basal ganglia signal	Drug history; MRI changes regress after withdrawal; CSF GABA
T1 Loop and thiazide diuretics surreptitious or unreported use	Barter · Gitelman syndrome reversible <i>SLC12A1 · SLC12A3</i> Renal	The same transporters, blocked pharmacologically	Urine diuretic screen; intermittent rather than stable biochemistry
T2 Aminoglycosides gentamicin, tobramycin	Antenatal Bartter syndrome reversible <i>CASR · BSND</i> Renal	CaSR activation suppresses thick ascending limb salt transport	Drug history; resolves on discontinuation

AGENT / EXPOSURE	MENDELIAN MIMIC	SHARED LESION	WHAT SEPARATES THEM
T1 Leflunomide, teriflunomide rheumatoid arthritis, multiple sclerosis	Miller syndrome (postaxial acrofacial dysostosis) <i>DHODH</i> Craniofacial	Dihydroorotate dehydrogenase inhibited	Maternal exposure; cholestyramine washout history; long drug half-life
T2 Warfarin, weeks 6–9 maternal anticoagulation	Chondrodysplasia punctata, X-linked (CDPX1) <i>ARSE</i> Skeletal	Vitamin K-dependent carboxylation of matrix Gla proteins fails	Maternal anticoagulation history; fetal sex; ARSE sequencing
T2 Benzocaine, dapsone, nitrites topical anaesthetics, well water	Congenital methaemoglobinaemia type I reversible <i>CYB5R3</i> Haematologic	Haemoglobin iron oxidised faster than the reductase recovers it	Reversible; NADH-cytochrome b5 reductase activity is normal
T1 Nitisinone (therapeutic) tyrosinaemia type I treatment	Tyrosinaemia type III · hawkinsinuria reversible <i>HPD</i> Metabolic	4-hydroxyphenylpyruvate dioxygenase inhibited	It is on the medication list; succinylacetone status

CLASS B

Transport, tubule and metal handling

6 pairs

Different node, same pathway or same organ compartment. Mostly nephrology and metal metabolism, where the acquired form is often the commoner cause of the classic genetic signature.

AGENT / EXPOSURE	MENDELIAN MIMIC	SHARED LESION	WHAT SEPARATES THEM
T2 Manganese welding fume, mining, ephedrone, long-term TPN	Hypermanganesaemia with dystonia reversible <i>SLC30A10</i> · <i>SLC39A14</i> Neurologic (CNS)	Mn accumulation in globus pallidus; T1 hyperintensity	Occupational or TPN history; polycythaemia, cirrhosis and childhood onset favour genetic
T2 Copper excess Indian childhood cirrhosis, idiopathic copper toxicosis	Wilson disease <i>ATP7B</i> Hepatic	Hepatic copper overload with Mallory bodies	Ceruloplasmin, Kayser-Fleischer rings, 24 h urinary Cu, ATP7B sequencing
T2 Zinc excess denture adhesive, supplements	Menkes disease (infants); B12-like myeloneuropathy (adults) reversible <i>ATP7A</i> Multisystem	Induced metallothionein blocks copper absorption	Serum Cu and ceruloplasmin recover on zinc withdrawal; no pili torti

AGENT / EXPOSURE	MENDELIAN MIMIC	SHARED LESION	WHAT SEPARATES THEM
T2 Cadmium, ifosfamide, tenofovir, valproate itai-itai disease; chemotherapy; antiretrovirals	Renal Fanconi syndrome reversible <i>CLCN5</i> · <i>OCRL</i> · <i>CTNS</i> Renal	Proximal tubular injury with generalised reabsorptive failure	Exposure history; leukocyte cystine; ocular and CNS findings
T3 Aristolochic acid herbal remedies; Balkan endemic nephropathy	Autosomal dominant tubulointerstitial kidney disease <i>UMOD</i> · <i>MUC1</i> Renal	Tubulointerstitial fibrosis with bland urine and minimal proteinuria	Herbal and geographic exposure; AA-DNA adducts; upper urothelial carcinoma
T2 Cisplatin, calcineurin inhibitors, amphotericin B oncology and transplant regimens	Gitelman syndrome reversible <i>SLC12A3</i> Renal	Distal tubular Mg ²⁺ and K ⁺ wasting	Drug history; partial recovery after discontinuation

CLASS C

Developmental and teratogenic

7 pairs

Exposure during a morphogenetic window reproduces a malformation syndrome. Timing is the whole story — the same agent outside the window does nothing.

AGENT / EXPOSURE	MENDELIAN MIMIC	SHARED LESION	WHAT SEPARATES THEM
T1 ACE inhibitors, ARBs, aliskiren maternal antihypertensives	Renal tubular dysgenesis <i>REN</i> · <i>AGT</i> · <i>ACE</i> · <i>AGTR1</i> Renal	Renin-angiotensin axis blocked during nephrogenesis	Maternal medication history across the whole pregnancy; consanguinity and recurrence favour genetic
T3 Mycophenolate mofetil maternal transplant immunosuppression	CHARGE syndrome <i>CHD7</i> Craniofacial	Uncertain; neural crest purine dependence	Maternal medication history; CHD7 negative; microtia and canal atresia pattern
T2 Isotretinoin acne therapy in pregnancy	22q11.2 deletion · oculo-auriculo-vertebral spectrum <i>TBX1</i> Cardiac	Retinoic acid signalling in cardiac neural crest	Maternal exposure; chromosomal microarray; thymic and parathyroid status
T2 Valproate (fetal) maternal antiepileptic therapy	Coffin-Siris and related BAFopathies <i>ARID1B</i> · <i>SMARCB1</i> Neurologic (CNS)	HDAC inhibition; chromatin remodelling disrupted	Maternal AED history; neural tube defect pattern; fifth-digit nail hypoplasia
T3 Thalidomide historic; ongoing use in leprosy and myeloma	Roberts/SC phocomelia · Holt-Oram · TAR <i>ESCO2</i> · <i>TBX5</i> · <i>RBM8A</i> Skeletal	Cereblon neosubstrate degradation in the limb bud	Era and geography; TAR spares thumbs; Holt-Oram is radial-ray with conduction disease

AGENT / EXPOSURE	MENDELIAN MIMIC	SHARED LESION	WHAT SEPARATES THEM
T3 Alcohol (FASD) prenatal exposure	Cornelia de Lange · Williams · 22q11.2 <i>NIPBL</i> · <i>ELN</i> Craniofacial	Multiple; craniofacial, growth and neurodevelopmental overlap	Gestalt overlap is high — microarray and targeted testing rather than clinical judgement alone
T3 Methotrexate, aminopterin maternal chemotherapy or abortifacient use	Craniosynostosis syndromes <i>FGFR2</i> · <i>FGFR3</i> Craniofacial	Folate antagonism during cranial and limb development	Exposure timing; suture pattern; limb findings

CLASS D

Storage, ultrastructure and imaging mimics

11 pairs

The trap is a biopsy or a scan. These are the cases where a pathologist or radiologist raises a genetic diagnosis and the medication list would have closed it.

AGENT / EXPOSURE	MENDELIAN MIMIC	SHARED LESION	WHAT SEPARATES THEM
T3 Amiodarone, hydroxychloroquine, chloroquine antiarrhythmic and antirheumatic therapy	Fabry disease reversible <i>GLA</i> Renal	Cationic amphiphilic drug phospholipidosis; lamellar inclusions	α -galactosidase A activity and lyso-Gb3; curvilinear vs zebra bodies; the drug list
T3 Chloroquine long-term antimalarial or antirheumatic use	Danon disease <i>LAMP2</i> Muscle	Autophagic vacuolar myopathy with cardiac involvement	LAMP2 immunostaining; X-linked pedigree; WPW pattern
T3 Pentosan polysulfate sodium interstitial cystitis, often for years	Pattern dystrophy · Stargardt disease <i>PRPH2</i> · <i>ABCA4</i> Ophthalmic	RPE toxicity with parafoveal pigmented deposits	Cumulative dose and duration; negative IRD panel; characteristic OCT and FAF
T3 Hydroxychloroquine cumulative dose over years	Bull's-eye maculopathy · ABCA4 retinopathy <i>ABCA4</i> Ophthalmic	Outer retinal and RPE toxicity	Cumulative dose; symmetry; ERG and inherited retinal dystrophy panel
T2 n-Hexane, 2,5-hexanedione, carbon disulfide glue sniffing, solvent industries	Giant axonal neuropathy reversible <i>GAN</i> Neurologic (PNS)	Neurofilament cross-linking; giant axons on nerve biopsy	Solvent exposure; kinky hair and childhood onset favour genetic
T1 Stavudine, zidovudine, linezolid antiretroviral and antibiotic therapy	mtDNA depletion syndromes reversible <i>POLG</i> · <i>TK2</i> Multisystem	Pol- γ inhibition; mitoribosome inhibition	Drug history; mtDNA copy number recovers off drug; lactate trajectory

AGENT / EXPOSURE	MENDELIAN MIMIC	SHARED LESION	WHAT SEPARATES THEM
T2 MPTP, rotenone, paraquat contaminated opioid analogues; agricultural exposure	Early-onset parkinsonism <i>PRKN</i> · <i>PINK1</i> Neurologic (CNS)	Complex I inhibition; nigrostriatal dopaminergic loss	Exposure history; DAT scan is abnormal in both; family history and levodopa response
T3 Carbon monoxide, methanol acute poisoning	Glutaric aciduria type 1 · Leigh syndrome <i>GCDH</i> · <i>SURF1</i> Neurologic (CNS)	Pallidal and putaminal necrosis on MRI	Acute onset; urine organic acids; carboxyhaemoglobin and osmolar gap
T3 Vinyl chloride polymerisation plant work	Hajdu-Cheney syndrome <i>NOTCH2</i> Skeletal	Acro-osteolysis with sclerodermatous skin and Raynaud	Occupational history; absent wormian bones and serpentine fibula
T3 Beryllium aerospace, dental alloy, electronics	Blau syndrome / early-onset sarcoidosis <i>NOD2</i> Immune	Non-caseating granulomatous inflammation	Beryllium lymphocyte proliferation test; occupational history; uveitis pattern
T3 Heroin pyrolysis vapour “chasing the dragon”	Vanishing white matter disease <i>EIF2B1</i> · <i>EIF2B5</i> Neurologic (CNS)	Spongiform leukoencephalopathy with symmetric white matter loss	Exposure history; acute onset; MRS lactate peak; no stress-triggered decline

CLASS E

Fibrosis and end-organ remodelling

10 pairs

Chronic injury converging on a fibrotic endpoint. Weakest mechanistically, and the most commonly missed, because the exposure is remote and the phenotype is slow.

AGENT / EXPOSURE	MENDELIAN MIMIC	SHARED LESION	WHAT SEPARATES THEM
T2 Valproate antiepileptic therapy in young children	Alpers-Huttenlocher / POLG hepatopathy <i>POLG</i> Hepatic	Mitochondrial hepatotoxicity; mtDNA depletion	POLG sequencing and liver mtDNA copy number — and sequence before prescribing
T3 Pyrrolizidine alkaloids bush tea, comfrey, Senecio, contaminated grain	Veno-occlusive disease with immunodeficiency <i>SP110</i> Hepatic	Sinusoidal obstruction; centrilobular necrosis and fibrosis	Herbal history; VODI has infantile onset with hypogammaglobulinaemia
T3 Chronic arsenic tube-well water, West Bengal and Bangladesh	Congenital hepatic fibrosis <i>PKHD1</i> Hepatic	Non-cirrhotic portal fibrosis with preserved hepatocellular function	Water arsenic, keratoses, Mees lines; renal cysts and ductal plate malformation favour genetic

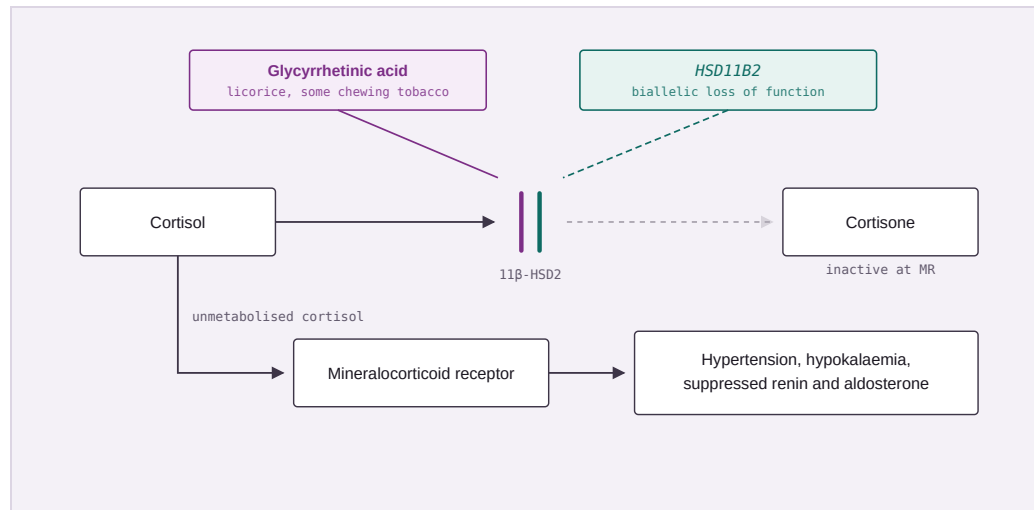
AGENT / EXPOSURE	MENDELIAN MIMIC	SHARED LESION	WHAT SEPARATES THEM
T3 Vinyl chloride polymerisation plant work	Congenital hepatic fibrosis / portal hypertension <i>PKHD1</i> Hepatic	Hepatoportal sclerosis; non-cirrhotic portal hypertension	Occupational history; angiosarcoma surveillance; coexisting acro-osteolysis
T3 Methotrexate, amiodarone, tamoxifen long-term therapy	Lysosomal acid lipase deficiency · hypobetalipoproteinaemia reversible <i>LIPA</i> · <i>APOB</i> Hepatic	Microvesicular steatosis progressing to fibrosis	Drug and cumulative-dose history; LAL enzyme activity; lipid profile
T2 Parenteral nutrition; anabolic androgens intestinal failure; performance use	PFIC · benign recurrent intrahepatic cholestasis reversible <i>ABCB11</i> · <i>ATP8B1</i> · <i>ABCB4</i> Hepatic	Canalicular bile salt export impaired	GGT pattern; nutrition and supplement history; resolves on withdrawal
T3 Cobalt metal-on-metal arthroplasty; historic beer additive	Mitochondrial and Friedreich-like cardiomyopathy <i>FXN</i> Cardiac	Impaired oxidative phosphorylation in myocardium	Serum cobalt and implant history; coexisting hypothyroidism and deafness
T3 Anthracyclines childhood and adult oncology	Dilated cardiomyopathy <i>TTN</i> · <i>LMNA</i> Cardiac	Topoisomerase-2β mediated myocyte loss	Treatment history — but truncating TTN variants genuinely raise risk, so both can be true
T3 Silica, cobalt-tungsten hard metal engineered stone, foundry, tool grinding	Familial pulmonary fibrosis · surfactant disorders <i>TERT</i> · <i>SFTPC</i> · <i>MUC5B</i> Pulmonary	Fibrotic interstitial remodelling	Occupational history and mineralogic analysis; telomere length; family history
T3 Benzene, alkylating chemotherapy solvent exposure; prior oncology treatment	Fanconi anaemia <i>FANCA</i> · <i>FANCC</i> Haematologic	Marrow failure with clonal evolution to MDS	Chromosome breakage testing (DEB/MMC); radial ray anomalies, short stature, café-au-lait

Four worked out

Chosen because the mechanism is drawable: in each, a chemical and a variant disable the same node, and the diagram is the argument.

01 Licorice and *HSD11B2*: one enzyme, two ways to lose it

ENDOCRINE · RENAL



Both lesions land on the same enzyme. Downstream of that node there is no biochemical difference to find — the urinary cortisol:cortisone ratio is elevated in either case.

Apparent mineralocorticoid excess exists because 11β-HSD2 is what stops cortisol — circulating at roughly a thousand times aldosterone's concentration — from occupying the mineralocorticoid receptor. Remove the enzyme and cortisol becomes a mineralocorticoid. Biallelic *HSD11B2* variants remove it genetically; glycyrrhetic acid, the active metabolite of licorice, removes it pharmacologically.

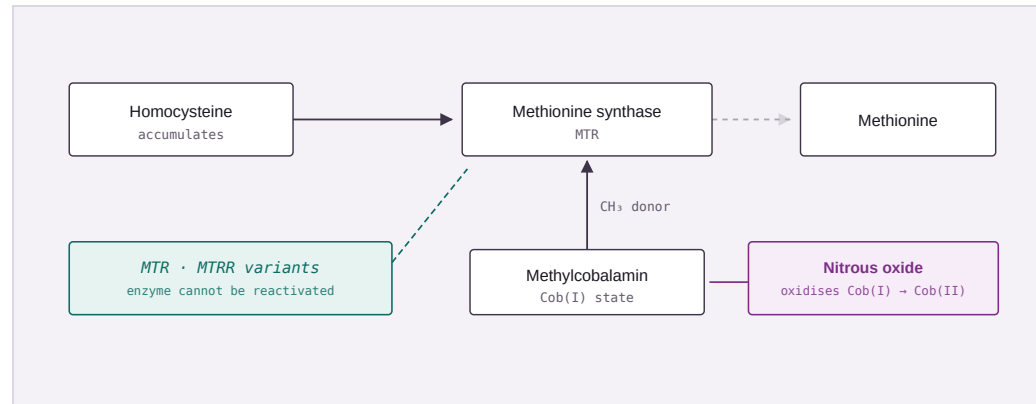
The presentation is the same on both routes: early hypertension, hypokalaemic metabolic alkalosis, low renin, low aldosterone, and an elevated urinary cortisol-to-cortisone ratio. Nothing in the biochemistry distinguishes them, because the biochemistry is the shared lesion.

WHAT BREAKS THE TIE

Dietary and product history — licorice confectionery, herbal teas, some smokeless tobacco. The acquired form normalises over weeks of abstinence; the genetic form does not. Age of onset and family history help, but withdrawal is the decisive test.

02 Nitrous oxide: a remethylation disorder you can buy at a corner shop

NEUROLOGIC · METABOLIC



The variants disable the enzyme; N₂O disables its cofactor. Different nodes, one functional deficit — and one indistinguishable metabolite profile.

Nitrous oxide irreversibly oxidises the cobalt centre of cobalamin, and methionine synthase cannot run without it. The result is a functional remethylation defect: total homocysteine rises, methylmalonic acid rises, and the patient develops subacute combined degeneration of the cord with a length-dependent neuropathy and, sometimes, psychiatric change.

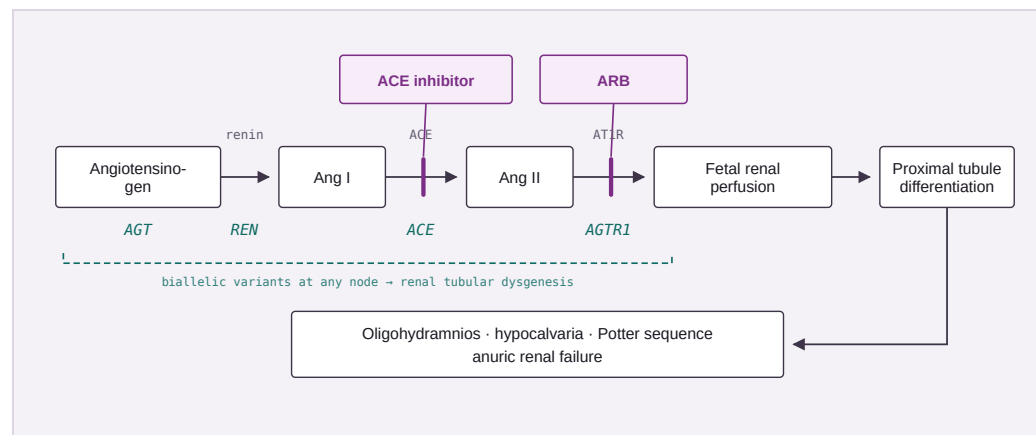
Sent to a metabolic clinic, that panel reads as an inborn error of cobalamin metabolism. With recreational N₂O now sold in litre canisters, this is the phenocopy on this page most likely to actually walk into a clinic — and adolescents and young adults are exactly the group in whom a late-onset cblG or cblE presentation would be plausible.

WHAT BREAKS THE TIE

Ask directly about nitrous oxide use; patients rarely volunteer it and often do not consider it a drug. Metabolites correct with hydroxocobalamin and abstinence within weeks. Genetic remethylation disorders do not fully normalise, and usually carry a longer history.

03 Fetal RAS blockade: four genes and two drug classes, one axis

RENAL · FETAL



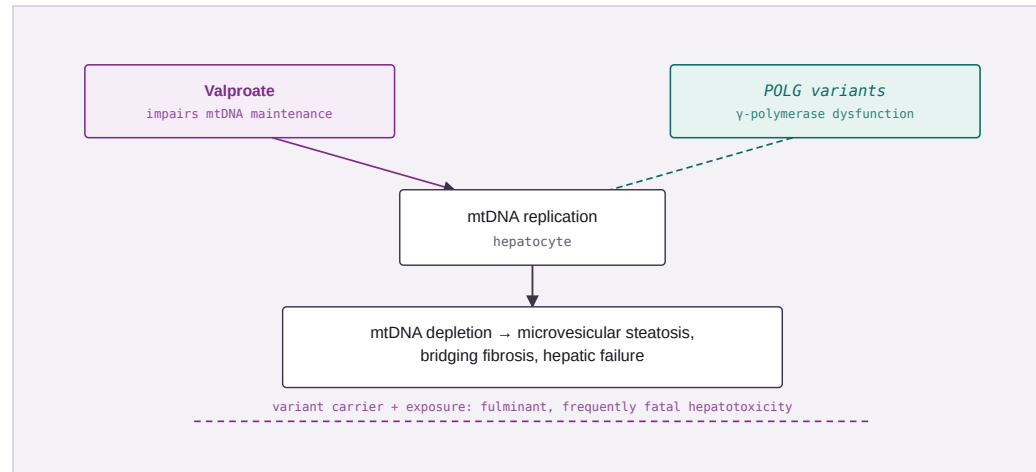
Autosomal recessive renal tubular dysgenesis is caused by variants at four different nodes of one axis. Maternal ACE inhibitors, ARBs or aliskiren interrupt the same axis pharmacologically — and the renal histology is the same: absent or short proximal tubules.

This is the purest illustration of why phenotype-driven prioritisation cannot save you. The phenotype is not caused by any particular gene; it is caused by the loss of angiotensin II signalling during nephrogenesis. Four genes and three drug classes are simply different ways to produce that loss.

Fetal exposure after the first trimester gives oligohydramnios, hypocalvaria from the resulting compression, pulmonary hypoplasia, and anuric renal failure at birth — the identical picture to recessive RTD, down to the histology on autopsy.

WHAT BREAKS THE TIE

Maternal medication reconciliation for the whole pregnancy, including antihypertensives started before the pregnancy was known. Consanguinity, a previously affected sibling, or exposure that stopped before week 12 all push back toward a genetic cause — and warrant sequencing the four genes rather than closing the case.



The only entry here where the two routes are not alternatives. Valproate phenocopies *POLG*-related hepatopathy in people with normal *POLG* — and precipitates it catastrophically in those without.

Valproate-induced hepatotoxicity produces microvesicular steatosis, lactic acidosis and progressive liver failure in a child who may also have refractory seizures. That is Alpers-Huttenlocher syndrome. A referral arriving as "intractable epilepsy plus liver failure" will be worked up for a mitochondrial hepatocerebral disorder, and often should be.

What makes this one worth carrying separately is that the relationship runs both ways. In children with biallelic *POLG* variants, valproate does not merely resemble the disease — it converts a survivable course into a fatal one. The phenocopy and the gene–environment interaction share a node, which is why *POLG* testing before valproate in unexplained childhood epilepsy has become standard practice in many centres.

WHAT BREAKS THE TIE

Liver mtDNA copy number and *POLG* sequencing, not the biopsy appearance. And in a child under two with unexplained epilepsy, the ordering of the question matters more than the answer: sequence before prescribing.

Why this matters for a phenocopy tab

None of these are solved by adding HPO terms. A richer phenotype profile makes the mimicked gene rank *higher*, because the phenotype genuinely matches — this is the failure mode where deep phenotyping and phenotype-driven prioritisation cooperate to produce a confident wrong answer.

What resolves them is orthogonal to the phenotype: occupational and medication history, dietary and herbal exposure, a tox screen, urine diuretics, blood metals, cumulative drug dose. Almost none of that is captured as structured data alongside HPO in a diagnostic pipeline. ECTO and ExO exist and see essentially no use in rare-disease workflows.

Tier 1 is also where an AOP linkage would be tightest and most mechanical: the molecular initiating event is the gene product, which gives a clean MIE ↔ gene join key rather than a fuzzy phenotype-level mapping.

Caveats on this compilation

- Tier assignments are a judgement about mechanistic tightness, **not a graded evidence rating**. They would need review before curation.
- Two entries (anthracyclines/*TTN*, silica/*TERT*) are arguably gene–environment interaction rather than phenocopy proper. Kept in and flagged, but they may belong in a separate category.
- Valproate/*POLG* is deliberately both: a phenocopy of Alpers-Huttenlocher in people with normal *POLG*, and a fatal trigger in those without. If the schema expresses one relation per pair, that case needs a second edge type.
- This is a literature compilation, not a systematically curated set. Treat it as a starting list, not a reference.

SELECTED SOURCES

NTE/PNPLA6 and organophosphate delayed neurotoxicity — PMC7271139 · Cyp51 knockout resembles Antley-Bixler syndrome — PMC3190716 · Aripiprazole and trazodone elevate 7-dehydrocholesterol — PMID 23628460 · Inherited manganese disorders — PMC8061520 · Renal tubular dysgenesis — OMIM 267430 · Hydroxychloroquine-induced mimic of renal Fabry disease — Kidney Int 2018 · Zebra bodies on renal biopsy: Fabry disease? — BMC Nephrol 2017 · Pentosan polysulfate maculopathy diagnosed as pattern dystrophy — PMID 34039907 · Pentosan polysulfate maculopathy diagnosed as Stargardt disease — PMC6997837 · Hypoglycin A: mechanism of Jamaican vomiting sickness — Science 1972 · Ackee fruit toxicity — StatPearls NBK431101